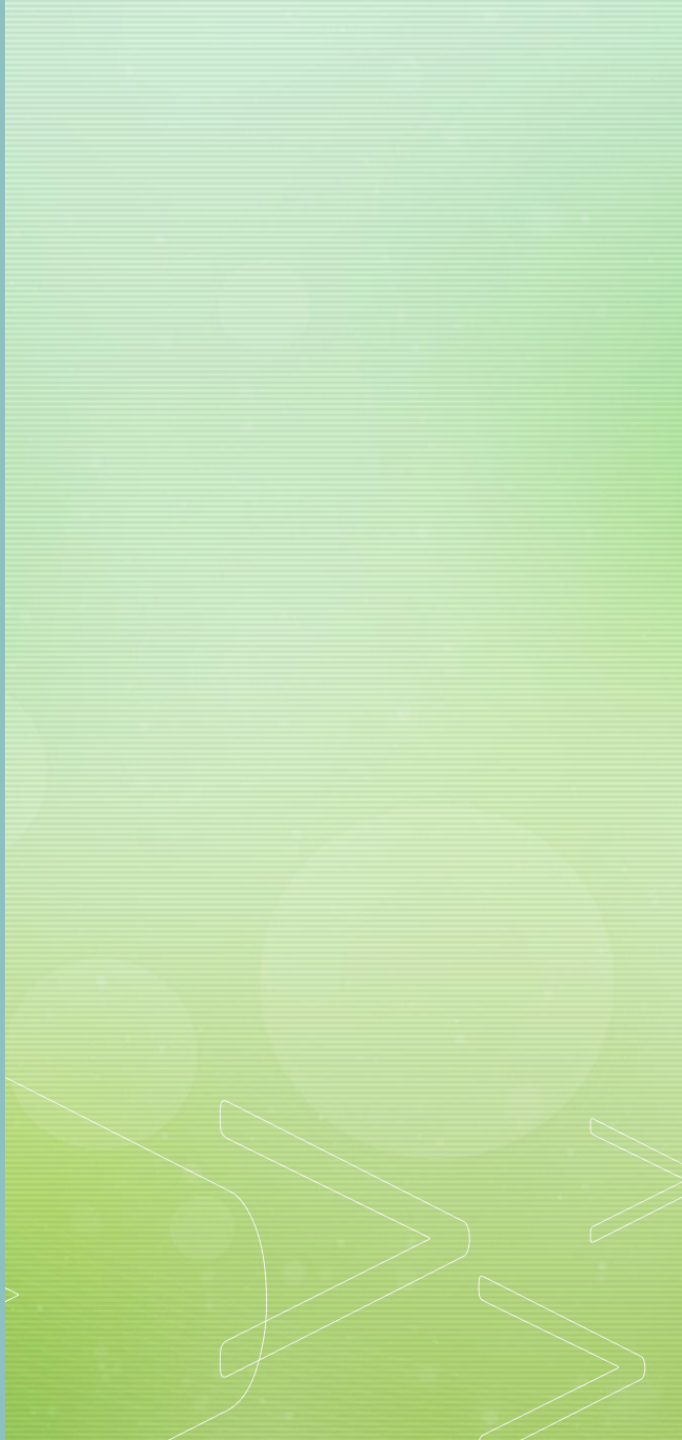




Unit 03: LITERATURE REVIEW

Scientific Research



LITERATURE REVIEW

- Doing a literature review is one of the first steps in a research project.
- That is to go through all the available information sources in order to **know the current state of knowledge** in the subject of the research project and to **assess it for relevance, quality, controversy and gaps**.
- A research project should try to resolve a controversy or to fill a gap.
- This lecture shows you where to find the necessary information and how to analyze and present it.

WHY REVIEW the LITERATURE?



“We are like dwarf on the shoulders of giants, so that we can see more than they, and things at a greater distance, not by virtue of any sharpness of sight on our pan, or any physical distinction, but because we are carried high and raised up by their giant size.”

JOHN OF SALISBURY, 12th century theologian and author

FUNCTIONS of a LITERATURE REVIEW

- Bringing clarity and focus to your research problem
 - By undertaking a literature review, you can understand the subject better and thus can **conceptualize your research problem clearly and precisely**.
 - When reviewing the literature you learn what aspects of your subject area have been examined by others, what they have found out about these aspects, what gaps they have identified and what suggestions they have made for further research.
 - Knowing these helps you **to focus your research on the gaps** in the current body of knowledge, thereby enhancing its relevance.

FUNCTIONS of a LITERATURE REVIEW

- Improving your research methodology
 - Familiarize you with the methodologies other scholars used to deal with similar research problems.
 - Know which procedures and methods have worked well and what difficulties have been faced with.
 - Be able to select a well-suited methodology for your research question.

FUNCTIONS of a LITERATURE REVIEW

- Broadening your knowledge base in your research area.
 - Know what other researchers have found in regard to the same or similar questions, and what theories have been put forward.
- Enabling you to contextualize your findings
 - Understand **how your findings fit into the existing body of knowledge.**
 - What contribution have you made?
 - How do your findings compare with others.

FUNCTIONS of a LITERATURE REVIEW

- Knowledge synthesis:
 - Integrate a set of ideas that were previously dispersed and turn them into a coherent framework
 - Clarify concepts that were only partially present in other works
 - Introduce a new / original (fresh) look into the subject
 - Show a critical perspective and some “personal touch” (how you see the current state of the art)

FUNCTIONS of a LITERATURE REVIEW

- The review is an important introduction to the research project and underpins the argument about why the project is worth doing.
- The literature review tends to reveal a lot about:
 - the attitude of the researcher
 - and the seriousness of his/her intentions,
 - as well as the level of organization and clarity of thought achieved.

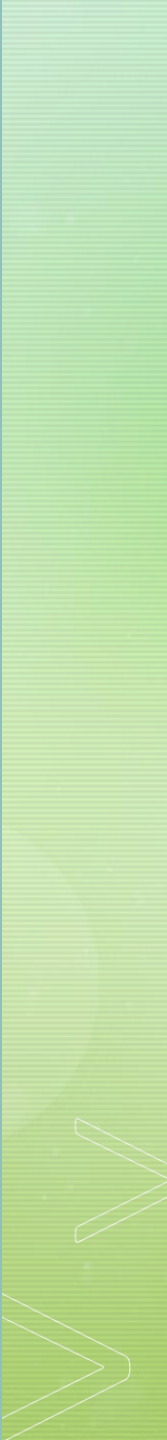
LITERATURE REVIEW SUBTASKS



1. *Carrying out a literature search*
2. *Reading the selected literature*
3. *Writing a literature review*

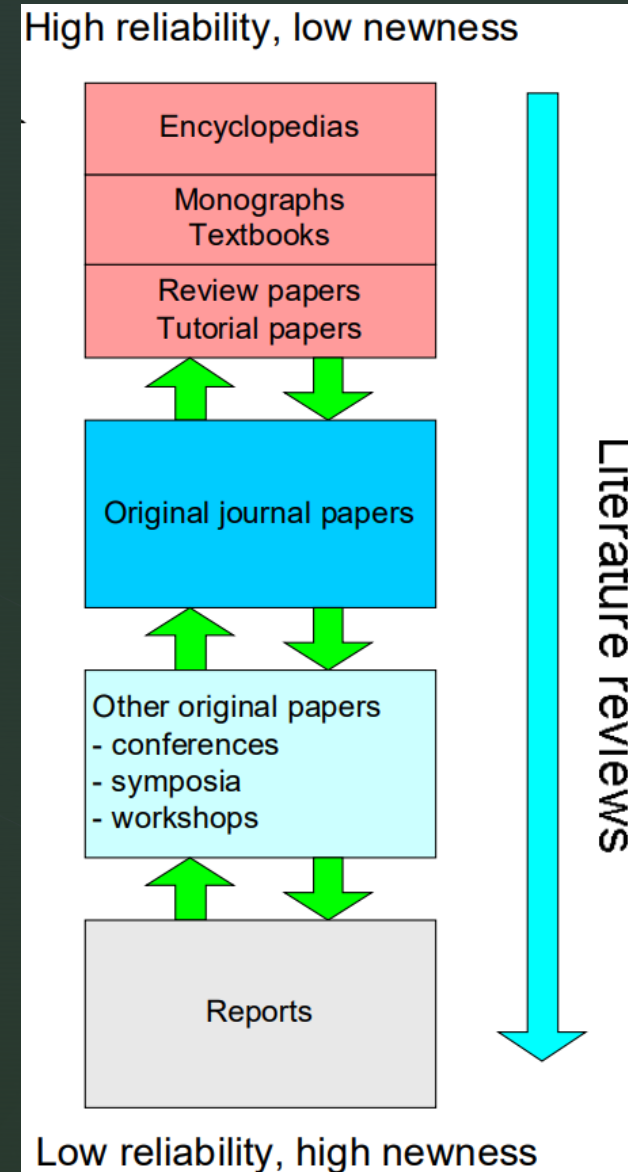


1. LITERATURE SEARCH



LITERATURE SEARCH

- Sources:
 - Books
 - Journal Papers
 - Conference Papers
 - Technical Reports



[Mämmelä, 2006]

LITERATURE SEARCH

- The literature that your research will rely on **should be from a refereed and reputable source**, such as:
 - papers that have been reviewed and published in a reputable venue,
 - works that have been undertaken and examined at a reputable institution,
 - and books that are based on the information presented in these.
- Other literature—news articles, science magazines, Wikipedia pages, or documentation, for example—may alert you to the existence of reputable work but is rarely worth citing.

WAYS of SEARCHING the LITERATURE

- Use obvious search terms to explore the Web.
 - You are likely to find, not just papers, but also home pages for projects and teams concerned with the same research area.
 - You are also likely to find documents that suggest further valuable search terms. Be exploratory in your search; sometimes the research in an area is divided across separate communities that have different vocabularies.
- Some of the major search engines have **search tools that are specifically for academic papers** (Google Scholar, Microsoft Academic Search). They are today the single most effective method for finding relevant work.

WAYS of SEARCHING the LITERATURE

- Visit the websites of research groups and researchers working in the area. These sites should give several kinds of links into the wider literature:
 - the names of researchers whose work you should investigate,
 - the names of their co-authors,
 - conferences where relevant work appears,
 - and papers with lists of references to explore.
- Follow up the references in promising research papers. These indicate relevant individuals, conferences, and journals.

WAYS of SEARCHING the LITERATURE

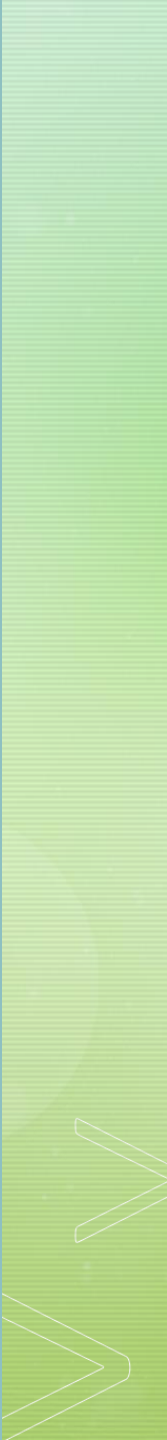
- Browse the recent issues of the journals and conferences in the area; search other journals and conferences that might carry relevant papers.
- Search the publisher-specific **digital libraries**.
 - These include publishers such as Wiley and Springer, and professional societies such as the ACM and IEEE.
 - There are also a wide range of online archives, in particular www.arXiv.org.

WAYS of SEARCHING the LITERATURE

- Most conferences have websites that list the program, that is, the papers to appear in the conference that year. Within a conference, papers are often grouped by topic—another hint of relevance.
- Discuss your work with as many people as possible. Some of them may well know of relevant work you haven't encountered.



2. CRITICAL READING



READING the SELECTED LITERATURE

- After having a list of selected papers, next step is to check which ones that you need to read.
- For each of the selected papers:
 - scan abstract
 - read introduction carefully
 - read conclusions quickly
 - look at the results
 - look at references
 - skim rest of paper
- ... and then go back and read the paper critically if it potentially contains something that is relevant to your research project.

READING the SELECTED LITERATURE

- You should take a broad definition of “relevant”, not just those papers that have proposals for competing methods.
 - Does the paper have interesting insights into other research literature?
 - Does it establish a benchmark?
 - Have the authors found a clever way of proving a theorem that you can apply in your own work?
 - Does the paper justify a decision to not pursue some particular line of investigation?
 - Other people’s research can have many different kinds of effect on your work.

READING the SELECTED LITERATURE

- Good researchers must have the ability to objectively analyze the work and claims of others.
 - With experience, you can place each paper in a context of other work that you know and assess it on a range of characteristics.
 - You will also become alert to common mistakes and bogus claims.
- Don't accept something as true just because it was published.

READING the SELECTED LITERATURE

- Be aware that much research is just misguided.
 - People investigate problems that are already solved and well understood
 - or don't realize that the proposed improvement actually makes the method worse.
 - mathematics may be pointless;
 - assumptions may be implausible;
 - evaluation strategies may not make sense.
 - The data set used may be so tiny that the results are meaningless.

READING the SELECTED LITERATURE

- When reading papers, you should **be questioning, balanced, and skeptical** because:
 - They are a presentation of new work rather than a considered explanation of well-known results
 - The constraints of writing to a deadline mean that mistakes are undiscovered and some issues unexplored.
 - Some aspects of older papers may be superseded or irrelevant or may rely on limited or technically outdated assumptions.

READING the SELECTED LITERATURE

- Read papers by asking critical questions of them, such as:
 - Is there a contribution? Is it significant? Is the contribution of interest?
 - Are the results correct? Is the appropriate literature discussed?
 - Does the methodology actually answer the initial question?
 - Are the proposals and results critically analyzed?
 - Are appropriate conclusions drawn from the results, or are there other possible interpretations?
 - Are all the technical details correct? Are they sensible?
 - Could the results be verified? Are there any serious ambiguities or inconsistencies?

READING the SELECTED LITERATURE

- Actively attempt to identify the contributions and shortcomings rather than simply reading from one end to the other.
- Scientific contributions can take on many forms:
 - Some papers offer new ideas;
 - others implement ideas and show how they work;
 - others bring previous ideas together and unite them under a novel framework.

READING the SELECTED LITERATURE

- Write down your analysis of the paper.
- Notice that detailed analysis can be difficult before you develop a mature perspective by undertaking your own work.
- Thus reviewing of literature should not stop when the investigation begins but continue alongside the research.



3. WRITING A LITERATURE REVIEW



WRITING a LITERATURE REVIEW



- A literature review is **a structured analysis** of a body of literature.
- The papers in the review should be grouped by topic, and critically discussed in a way that allows the reader to understand their contribution to the field, their limitations, and the questions that they leave open.
- It is a good practice to **develop the review progressively**.

WRITING a LITERATURE REVIEW

- Begin a rough literature review as soon as you start reading.
- When you read a paper that you think will need to be discussed, add it in. As you add papers:
 - group them by topic and contribution,
 - add notes on each paper and how they relate to each other,
 - summarize each paper's contribution and the evidence used to support the claims,
 - and also note any shortcomings or features that are of interest.
- At this stage you should focus on organization and content rather than on presentation. You should focus on rewriting, editing, and polishing to produce the final literature review only when your research is complete.

WRITING a LITERATURE REVIEW

- As your review proceeds, it will become easier to decide whether to include each of the papers you read. Many factors will guide your decisions:
 - How close some other work is to yours, or how influential it has been.
 - You may find a survey paper, or a recent paper with a thorough literature review of its own, that means that many of the older papers do not need to be discussed;
 - Some papers that initially seemed important might on reflection seem less relevant, and can be set aside.
 - While a paper that at first seemed too theoretical or abstract may on further investigation be revealed as foundational.

WRITING a LITERATURE REVIEW



- In the final literature review, you should:
 - Justify criteria for inclusion and exclusion of literature from review.
 - Distinguish what has been done in the field from what needs to be done.
 - Place the topic or problem in the broader scholarly literature.
 - Place the research in the historical context of the field

WRITING a LITERATURE REVIEW



- In the final literature review, you should:
 - Acquire and enhance the subject vocabulary
 - Articulate important variables and phenomena relevant to the topic
 - Identify the main methodologies and research techniques that have been used in the field, and their advantages and disadvantages.

WRITING a LITERATURE REVIEW



- In the final literature review, you should:
 - Synthesize and gain a new perspective on the literature
 - Related ideas and theories in the field to research methodologies
 - Rationalize the practical significance of the research problem
 - Be written with a coherent, clear structure that supported the review

LITERATURE REVIEW & Research Evaluation

- Towards the end of your thesis/dissertation/report [or paper] you will refer back to literature review
 - Do your findings **confirm** those of others?
 - Does your work **extend** that of others?
 - Does your work **provide new meaning** to the work of others?
 - Does your work **break new ground**?
 - Does your work **raise issues about the methodological choices** made in previous studies?
 - Does your work **challenge existing ideas** on your subject?

REFERENCE and CITATION



- All the theses, papers, and books that you discussed in the review need to be cited according to a standardized format.
- There are 3 major citation styles used in academic writing:
 - Modern Language Association (MLA)
 - American Psychological Association (APA)
 - Chicago style.
- Conferences and journals usually provide their own style.

REFERENCE and CITATION



- Don't cite to support common knowledge. For example, use of a binary tree in an algorithm doesn't require a reference to a data structures text.
- But claims, statements of fact, and discussion of previous work should be substantiated by reference

REFERENCE and CITATION

- On rare occasions it is necessary to refer to a result in an inaccessible paper.
 - For example, suppose that in 1981 Dawson wrote “Kelly (1959) shows that stable graphs are closed”, but Kelly (1959) is inaccessible and Dawson (1981) does not give the details.
 - In your write-up, do not refer directly to Kelly—after all, you can’t check the details yourself, and Dawson may have made a mistake.
 - ✓ *According to Dawson (1981), stable graphs have been shown to be closed.*
 - ✓ *According to Kelly (1959; as quoted by Dawson 1981), stable graphs are closed.*

REFERENCE and CITATION

- Cited material often uses different terminology, spelling, or notation, or is written for an entirely different context. When you use results from other papers, be sure to show the relationship to your own work.
 - For example, a reference might show a general case, but you use a special case; then you need to show that it is a special case.
 - If you claim that concepts are equivalent, ensure that the equivalence is clear to the reader.

REFERENCE and CITATION

- References that are discussed should not be anonymous.

- ✗ *Other work [16] has used an approach in which...*

- ✓ *Marsden [16] has used an approach in which...*

- Other work (Marsden 1991) has used an approach in which*

The modified versions provide more information to the reader, and “Marsden” is easier to remember than “[16]” if the same paper is discussed later on.

- Likewise, self-references should not be anonymous—it should be clear to the reader that references used to support your argument are your own papers, not independent authorities.

- ✗ *Smith et al. [10] found compressed lists to be...*

- ✓ *In Smith et al. [10], we found compressed lists to be...*

Literature Review CHECKLIST



1. Have you shaped your LR to fit your research questions or hypotheses?
2. Have you struck an appropriate balance between description and evaluation?
3. Have you sufficiently explained why certain sources were included or excluded?
4. Are there possibly any "missing inputs" that your readers or reviewers are likely to pick up on? In other words have you adequately covered the research territory?
5. Have you described the literature in an original manner so that questions of plagiarism will not arise?

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